Behaviour & Information Technology

ISSN: 0144-929X (Print) 1362-3001 (Online) Journal homepage: www.tandfonline.com/journals/tbit20

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To cite this article: Nur Efsan Cetinkaya & Nicole Krämer (2026) Between transparency and trust: identifying key factors in AI system perception, Behaviour & Information Technology, 45:5, 840-854, DOI: 10.1080/0144929X.2025.2533358

To link to this article: https://doi.org/10.1080/0144929X.2025.2533358

BEHAVIOUR & INFORMATION TECHNOLOGY
2026, VOL. 45, NO. 5, 840–854
https://doi.org/10.1080/0144929X.2025.2533358

Between transparency and trust: identifying key factors in AI system perception

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ABSTRACT

With the deployment of AI systems across multiple domains, understanding how users develop trust has becomecrucial for successful implementation. This study investigates how different AI features influence the decision to use an AI system and which characteristics users prioritise when evaluating them. We focus on whether users prefer systems whose functioning they can understand or whose trustworthiness is certified. We examined whether users favour system transparency through explainability features or rely more on external trust signals, such as AI certification seals, while considering how these preferences interact with technical reliability and fairness. Using conjoint analysis, we systematically compared the influence of four key attributes (transparency by explainability features, technical reliability, external trust signals through AI certifications, and fairness) on user decisions to use an AI system. Through cluster analysis, we identified two groups with opposing preferences and demographic differences. The first group prioritised high explainability and strong AI certification while showing negative preferences for fairness, whereas the second group favoured fairness and reliability while displaying negative attitudes toward explainability and AI certification. These contrasting prioritisation patterns raise important questions about AI systems development, particularly regarding challenges of addressing competing user requirements for trust-related features.

ARTICLE HISTORY
Received 25 April 2025
Accepted 7 July 2025

KEYWORDS
Artificial intelligence; XAI; trust in AI; reliability; fairness

1 Introduction

The use of artificial intelligence (AI) has grown rapidly in recent years, fundamentally transforming various aspects of society. As AI becomes more deeply integrated into daily life, understanding how users evaluate and choose to adopt AI systems becomes increasingly critical. Research has extensively investigated various attributes that influence AI adoption and trust, examining factors such as system explainability (Wing 2021), reliability (Ryan 2020), and fairness (Varona and Suárez 2022). While these individual studies have provided valuable insights, they predominantly examine each attribute in isolation, leaving a critical gap in our understanding of how these attributes interact and compare in importance when users make actual adoption decisions.

This gap is particularly significant as users in practice must evaluate multiple attributes simultaneously, not in isolation. When deciding whether to adopt an AI system in high-stakes scenarios, users cannot consider explainability without also weighing factors like reliability and fairness. Moreover, with the emergence of trust signals such as AI certification seals (Scharowski et al. 2023), users now have additional external indicators to consider alongside system attributes. This creates a complex decision environment where understanding the relative importance of each attribute becomes crucial for both theory and practice.

The relationship between understanding and trust in AI systems presents a fundamental tension in user adoption decisions. While some researchers argue that understanding AI systems through explainability leads to increased user (Shin 2021; Weitz et al. 2019), others suggest that external trust mechanisms may be equally or more important (Krämer, Wischnewski, and Müller 2023). The latter assumption is based on the observation that users might show unwillingness (Ngo and Krämer 2021) or inability (Bromme and Gierth 2021) to understand complex functioning and issues. This leads to the important question whether users prefer to understand AI systems directly, or whether they are willing to rely more heavily on external validations of trustworthiness.

Research has identified several key attributes that influence AI adoption. Explainability enables users to understand system decisions (Sheth et al. 2021), while

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reliability ensures consistent performance under specific conditions (Hong et al. 2022). External trust signals, such as AI seals of approval, offer third-party validation of system quality (Scharowski et al. 2023), particularly valuable for non-expert users (LaRose and Rifon 2006). Fairness, defined as the equitable treatment of diverse user groups, has emerged as another critical consideration as AI systems increasingly make decisions that affect different populations. The growing awareness of algorithmic bias and its potential societal impacts has made fairness a key factor in users' evaluation of AI systems (Bartneck et al. 2021). Each of these attributes potentially influences users' willingness to adopt AI systems, yet their relative importance remains unclear.

To address these questions, our study employs conjoint analysis to examine how users weigh different AI system attributes when making adoption decisions in high-stakes scenarios. This methodological approach is particularly well-suited for our research as it requires participants to make realistic trade-offs between different system attributes, revealing their true preferences more accurately than direct questioning. While our study does not directly measure trust, it operates on the premise that user preferences for these attributes indicate what users value most when deciding whether to adopt an AI system.

Our research focuses particularly on the interplay between understanding and trust, whether users prefer to understand AI systems directly through explainability, or whether they rely more heavily on external trust signals such as certifications. By examining these factors alongside reliability and fairness, we aim to provide insights that can guide both theoretical development and practical implementation of AI systems. For organisations developing AI systems for critical applications, understanding which attributes most strongly influence user acceptance can guide resource allocation and development priorities. For researchers, insights into how these attributes interact can inform more comprehensive models of AI adoption and trust.

2 Theoretical background

Given the increasing use of artificial intelligence (AI) systems in decision-making processes, there is a growing need to understand how users develop trust in these systems as well as what is affecting their adoption decision.

2.1 Trust development in AI systems

Trust and trustworthiness represent fundamental yet distinct concepts in the context of artificial intelligence adoption. While trust encompasses users' subjective perceptions of a system's reliability, trustworthiness refers to the system's objective characteristics, including its technical capabilities and performance (Schlicker and Langer 2021). This distinction is crucial for understanding how users interact with and accept AI technologies across various fields.

It is also important to distinguish between trust and reliance, as these concepts, while related, are not identical. In accordance with the findings of Lee and See (Lee and See 2004), trust can be defined as users' subjective perception and attitude towards a system's reliability. In contrast, reliance can be defined as the actual behavioural outcome of adopting or using the system. This distinction is crucial to the present study, as it examines users' choices between different AI systems (a form of intended reliance) rather than directly measuring their trust attitudes. While trust is known to influence reliance, other factors may also affect the decision to use a system. This highlights the importance of considering both concepts when examining AI adoption (Lee and See 2004).

The theoretical foundation for understanding trust in AI systems largely derives from Mayer et al.'s (Mayer, Davis, and David Schoorman 1995) model of organisational trust, which posits that trust develops based on a trustor's perceptions of a trustee's ability, benevolence, and integrity. Recognising that this organisational model may not perfectly translate to human-machine interactions (Madhavan and Wiegmann 2007), Lee and See (Lee and See 2004) adapted it specifically for automation contexts. In their framework, trust assessment relies on three key aspects: the machine's reliability and functionality (ability = performance), the intentions behind its design (benevolence = purpose), and its intelligibility (integrity = process). They conceptualise trust in automation as the belief that a machine will help achieve one's objectives, particularly in situations involving uncertainty and risk.

Building on this foundation, research has explored various aspects of trust formation in technological systems. One important consideration is epistemic trust, which represents the confidence users place in experts or institutions based on their perceived competence and integrity (Sperber et al. 2010). This concept highlights how trust in AI systems can be influenced by users' faith in the system's developers and implementing organisations. Another crucial aspect in trust development is the concept of calibrated trust, which examines the alignment between users' trust levels and a system's actual reliability (Wischnewski, Krämer, and Müller 2023). This alignment is essential for ensuring appropriate reliance on AI systems, as both over-trust and under-trust can lead to suboptimal outcomes.

Recent research has further differentiated between understanding and trust as distinct outcomes in human-AI interaction (De Brito Duarte et al. 2023). Understanding refers to comprehending the operational mechanisms of AI systems, while trust represents confidence in the system's outputs. This distinction is particularly relevant when examining the roles of explainability and trust seals in fostering user trust.

2.2 Understanding versus trust: explainability and trust seals

Explainability and trust seals represent two fundamentally different approaches to building trust in AI systems. Recently, and in line with considerable efforts of the explainable AI (XAI) community (Das and Rad 2020; Gunning et al. 2019; Langer et al. 2021; Norkute et al. 2021; Saeed and Omlin 2023), explainability, defined as the ability to attribute comprehensible and interpretable reasons for an AI's decisions and actions, has been widely recognised as an essential prerequisite for building user trust. It provides users with a deeper understanding of AI behaviour and reduces perceived risks and uncertainties (Shin and Park 2019). A significant body of research highlights that explainability is crucial for establishing AI trustworthiness (Caspers 2021; Jacovi et al. 2021).

However, research regarding the relationship between explainability and trust has yielded complex and sometimes contradictory findings. While some studies demonstrate that human-understandable explanations can enhance user comprehension and increase trust in AI performance (Nourani et al. 2019; Wang, Pynadath, and Hill 2015), others reveal a more nuanced dynamic. Ferrario and Loi (Ferrario and Loi 2022) found that enhancing system comprehension through explainability may actually result in decreased trust, as users gain deeper insights into system limitations and potential failure modes, leading to a more critical perspective of the system's capabilities. In addition, the attempt to foster a better understanding of system capabilities might be met with people's *unwillingness* or *inability* to understand (Krämer, Wischnewski, and Müller 2023). Here, it has been demonstrated that users sometimes do not want to understand algorithmic systems as they, for example, avoid to be scared off their usage (*unwillingness*, [Ngo and Krämer 2021]). Also, systems are increasingly complex to understand and merely a small part of everyday life problems that might be important to understand (e.g. climate change, vaccines), so that people are not able to achieve sufficient knowledge in all relevant socio-scientific issues (*inability*, [Bromme and Gierth 2021]).

Instead of building on understanding, Bromme and Gierth (Bromme and Gierth 2021) propose to shift the (too complicated) question of what to believe or to know, respectively, to the less demanding but equally rational question of whom to trust. This form of 'epistemic trust' (Sperber et al. 2010) is exemplified in trust seals. Trust seals, therefore, offer an alternative pathway to building trust through institutional validation rather than direct understanding. These trust cues, which include certifications and seals of approval from reputable organisations, provide users with tangible signals of an AI system's credibility and security (Scharowski et al. 2023). AI certifications represent formal recognition from independent third parties that a system meets predefined standards through thorough evaluation and auditing. These certifications aim to address the inherent complexity and uncertainty of AI systems by providing clear indicators of trustworthiness (Wischnewski, Krämer, and Müller 2023).

The effectiveness of trust seals is supported by research suggesting that public trust primarily requires robust regulatory oversight rather than technical understanding (Knowles and Richards 2021). This includes the establishment of authoritative bodies to enforce compliance with ethical standards and validate AI systems' trustworthiness through mandatory conformity assessments and algorithmic auditing (Afroogh et al. 2024). This approach acknowledges that users often rely more effectively on institutional validation ('whom to trust') than personal understanding ('what to know') when confronting complex socio-scientific issues (Bromme and Gierth 2021).

Recent research has begun to differentiate between understanding and trust as distinct outcomes in human-AI interaction (De Brito Duarte et al. 2023). While understanding involves comprehending operational mechanisms, trust represents confidence in system outputs. The influence of explainability on these outcomes varies: detailed technical explanations can enhance understanding while potentially reducing blind trust, leading to more informed and appropriate levels of reliance on the system (Mehrotra et al. 2024). This distinction helps explain why increased explainability does not automatically translate to increased trust and suggests that the goal should be to foster informed trust rather than maximising trust indiscriminately.

Despite extensive research on both mechanisms, there remains a significant gap in understanding how users weigh explainability against trust seals when evaluating AI systems. This study addresses this gap by directly comparing these approaches while considering additional influential factors.

2.3 Additional factors: reliability and fairness

While our primary focus remains on the interplay between explainability and trust seals, we include reliability and fairness as critical additional factors to provide a more comprehensive understanding of trust-building mechanisms in AI systems.

Reliability is defined as the probability that a system will fulfill its desired function under specific conditions over a designated period (Ebeling 2019). In the literature, reliability and robustness are often used interchangeably, though reliability specifically emphasises the temporal aspect of system performance (Hong et al. 2022). The concept has emerged as a crucial factor in ensuring the security and accuracy of AI-based systems (Belgaum et al. 2021), with reliability being closely intertwined with performance metrics, particularly accuracy (Mishra et al. 2024). Research shows a strong correlation between reliability levels and user trust and adoption rates (Kaplan et al. 2023). In this study, reliability therefore is included as a potential 'gold standard' for selecting a system to use.

Fairness, albeit a newer consideration in AI system development, has gained significant attention due to its implications for user trust and system adoption. Research demonstrates that users' trust in AI systems is significantly influenced by their perceptions of algorithmic fairness (Sullivan, De Bourmont, and Dunaway 2022). When AI systems demonstrate equitable treatment across different demographic groups, users are more likely to develop and maintain trust in these technologies. Conversely, instances of algorithmic discrimination can severely undermine user trust and impede AI adoption (Zhou et al. 2021). Algorithmic bias can manifest in various forms, often stemming from historical inequities in training data or inherent flaws in algorithm design (Jain and Menon 2023). For example, studies have documented cases where AI systems exhibited discriminatory patterns in healthcare diagnostics (Obermeyer et al. 2019) and hiring decisions (Peña et al. 2020). Users who encounter or become aware of such biases are significantly less likely to trust and engage with AI systems, regardless of their technical capabilities (Marassi 2023).

The inclusion of these additional factors allows us to evaluate how users prioritise different trust-building mechanisms while acknowledging that reliability and fairness play important roles in the broader context of AI system adoption. This approach enables us to better understand the relative importance of our primary variables – explainability and trust seals – within the larger landscape of factors influencing user trust in AI systems.

2.4 Hypotheses and research questions

Building on this theoretical framework, we identify several key research gaps regarding how users prioritise different trust-building mechanisms in AI systems. While extensive research exists on individual factors, there is limited understanding of how users weigh these factors against each other, particularly in the context of explainability versus institutional validation through trust seals.

A study has shown that fairness and performance are equally important to respondents whereas explainability is slightly less important (Kieslich, Keller, and Starke 2022). These findings also show the importance of reliability, since reliability is often categorised as a performance related factor (Kaplan et al. 2023). Furthermore, Kaplan et al. (Kaplan et al. 2023) demonstrated in their meta-analysis of trust in AI systems, reliability was identified as one of the most heavily weighted factors concerning the AI component. This is particularly evident in high-risk decision-making scenarios, where reliability emerges as the primary determinant of system adoption (Chancey et al. 2017). Therefore, the following hypothesis is proposed:

> H1: People will prefer reliability over fairness and explainability.

The relationship between explainability and trust seals presents a particularly interesting area for further examination. Explainability helps people to gain a deeper understanding of the decision-making process of an AI system (Zhou, Chen, and Holzinger 2022). In contrast, trust seals represent an external seal of approval that validates trust in the system without requiring understanding of its internal mechanisms (Cremers et al. 2019; Paaß and Hecker 2020). This distinction raises an important question about whether people prefer to understand the functioning of the system or whether they are ready to trust a system which has been certified as safe and trustworthy by a third party.

Building on the previously discussed framework of epistemic trust, research has shown that human understanding of algorithmic functioning is limited, with users frequently developing only superficial mental models of AI systems (DeVito et al. 2018; Kunkel et al. 2021; Ngo et al. 2020). While explanations might contribute to understanding, trust seals could potentially offer a more direct path to achieving calibrated trust, particularly when issued by competent institutions perceived as both honest and well-meaning (Hendriks, Kienhues, and Bromme 2015). This is especially important given that research has shown that providing extensive explanations can sometimes decrease users' ability to detect and correct system mistakes (Poursabzi-Sangdeh et al. 2021). Furthermore, studies have shown that users sometimes actively avoid detailed system information, fearing that increased knowledge about data processing might discourage future use (Ngo and Krämer 2021; Springer and Whittaker 2020). This behavioural pattern, combined with the cognitive demands of processing complex technical information, suggests that trust seals might provide a more accessible means of fostering appropriate trust levels. Based on these considerations, we hypothesise that people will prefer trust seals over explanations when establishing trust in AI systems. Therefore, the following hypothesis is proposed:

H2: People will prefer trust seal over explanations.

However, for the AI system to be trusted by the users, the AI's trustworthiness must be truly perceived by them. This requires certain cues to be provided to the users, which could be achieved through proper documentation. Therefore, other non-technical axiological factors for building trust, especially human-related ones, could be engineered to enhance trust without the need to improve the trustworthiness of AI (Afroogh et al. 2024). These axiological factors could be trust cues in the form of AI seals and also enabling fair and unbiased AI systems. Therefore, the following hypothesis is proposed:

H3: People will prefer communicated trustworthiness, enhanced through proper documentation (AI seal), over actual technical trustworthiness (reliability and fairness).

The degree of trust placed in AI is not a simple, universal phenomenon. Rather, it is a multifaceted and context-specific construct (Chen 2021). It is possible that users may assign differential weights to different attributes, depending on the specific application and their individual concerns (Dorton and Harper 2022). For instance, in the context of healthcare, explainability and performance are of paramount importance, whereas in customer service, user satisfaction may be the overarching concern (Geng and Chu 2012; Markus, Kors, and Rijnbeek 2021; Pierce et al. 2022).

The degree of trust placed in AI is influenced by a number of factors, including the explainability, reliability, trust cues like AI seals and fairness of the technology in question. Each attribute contributes to overall trust in a distinct manner (Angerschmid et al. 2022; De Brito Duarte et al. 2023; Dorton and Harper 2022; Shin 2021; Wischnewski, Krämer, and Müller 2023). The interaction between attributes can be complex. For example, high explainability might compensate for lower reliability or might mitigate concerns about fairness. It can be concluded that certain attributes may exert a more profound influence on trust than others. For instance, reliability is frequently regarded as a foundational attribute (Hong et al. 2022), yet fairness can also assume an important role contingent on the context. An understanding of the relative influence of different attributes can assist AI developers in directing their efforts towards the most impactful areas. For instance, if it is demonstrated that performance and fairness significantly enhance trust, designers may prioritise these features, even if achieving absolute reliability is challenging. This research question emphasises the necessity of a user-centred approach to the design of AI systems, whereby user trust considerations should guide the development process. It also aims to identify which individual attributes or combinations thereof have the most significant impact on trust. Therefore, we propose the following research question:

RQ: which combinations of attributes will be preferred the most?

2.5. RQ: which combinations of attributes will be preferred the most?

3 Method

The study design as well as the hypotheses were preregistered on the Open Science Framework (OSF) prior to data collection in November 2024 (https://osf.io/mw4dq/?view_only = 04327e74dfa34e4ca2babc170e86c048). All study materials, including instructions and data, are publicly available in the associated project repository. An approval by the responsible ethics committee of the University Duisburg-Essen was given to conduct the study. Statistical data analysis was performed using version 29 of IBM SPSS Statistics software for Windows (IBM Corp., 2023) and Python (Version 3.9.6)

3.1 Measures

3.1.1 Choice-Based conjoint analysis

The present study employed choice-based conjoint analysis (CBC) to systematically evaluate participants' trust preferences between different AI system configurations in healthcare decision-making. The CBC is a method of presenting choice scenarios to participants, who are required to make trade-offs between attributes. The method has been developed for the purpose of identifying true preference structures and determining the relative importance attributed to distinct characteristics. Despite its apparent simplicity, CBC is an effective method of capturing real-world decision processes by forcing users to make concrete trade-offs rather than rating features in isolation (Orme 2010). Prior to commencing the choice tasks, participants were presented with comprehensive instructions and a healthcare decision scenario that framed their evaluation specifically in terms of trust. The scenario described an AI system designed to assist with important health decisions, including treatment selection, health data monitoring, and personalised health advice. Participants were instructed to deliberate on which AI system profile they would entrust with greater confidence, contemplating the potential hazards of imprecise or prejudiced recommendations within healthcare contexts. The experimental design incorporated four key attributes with varying levels: explainability (three levels: no explainability, low explainability, high explainability), reliability (two levels: medium reliability at 65%, high reliability at 99%), trust cues (three levels: no AI seal, low AI seal, high AI seal), and fairness (two levels: low fairness, high fairness). To ensure a comprehensive understanding of the experimental design, participants received detailed descriptions of each attribute and its associated levels, with definitions readily available throughout the study for reference. The design generated 36 unique attribute combinations, which were presented across 18 discrete choice tasks (i.e. every participant had to make 18 choices between to system descriptions each). In each task, participants were shown two competing AI system profiles simultaneously and were required to indicate which system they would trust more based on the presented attributes (see Figure 1). This forced-choice methodology enabled the assessment of how different attribute combinations influenced participants' trust decisions. The systematic variation of attribute levels across profiles enabled the subsequent estimation of part-worth utilities and relative importance weights for each attribute and level in trust formation.

3.1.2 Additional measures

Attitude towards Artificial Intelligence was measured using the German version of the Attitude Towards Artificial Intelligence (ATAI) scale developed by Sindermann et al. (Sindermann et al. 2021). The scale consists of five items assessing two dimensions: Acceptance (2 items) and Fear (3 items) of Artificial Intelligence (e.g. 'Artificial intelligence will benefit humankind'). Participants responded on an 11-point Likert scale ranging from 0 ('strongly disagree') to 10 ('strongly agree'). The subscales demonstrated acceptable reliability in our sample, with Cronbach's $\alpha = .66$ for Acceptance and $\alpha = .70$ for Fear.

3.2 Procedure

The online study was conducted on the SoSci-Survey platform. Prior to the commencement of the study, participants were provided with a detailed instruction and asked to give consent. Following this, consent to participate was obtained. Participants then completed a choice-based conjoint analysis to assess their trust preferences in AI healthcare systems. Following this, they responded to a questionnaire measuring their attitudes toward artificial intelligence, specifically focusing on AI acceptance and fear of AI. The survey concluded with the collection of socio-demographic information, including gender, age, educational qualifications, and employment status. Upon completion of all study components, participants received a debriefing that outlined the study's objectives and purpose. The entire procedure

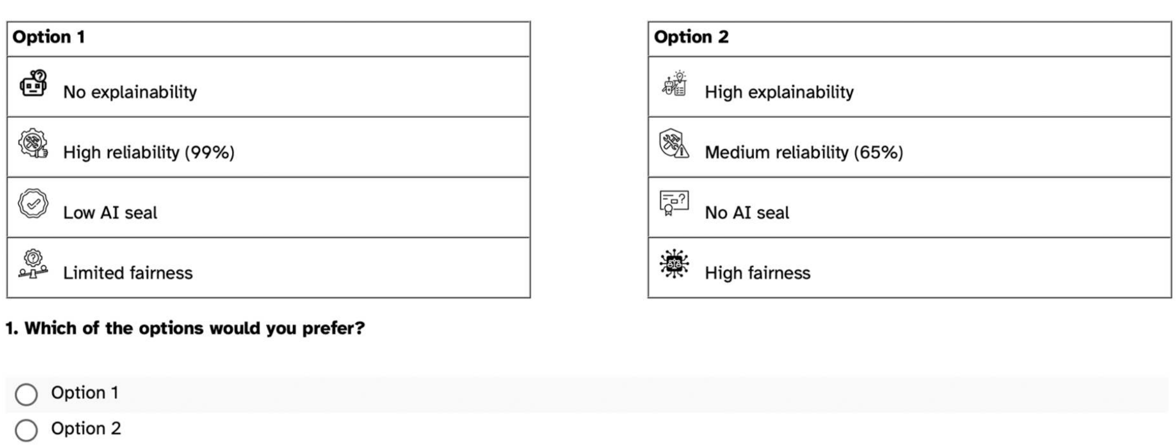

Figure 1. Example of a choice task in the conjoint analysis study.

was conducted in German to ensure participants could fully understand and respond to all materials in their native language.

3.3 Sample

The data collection process was carried out through an online study implemented on the SoSci-Survey platform. The required sample size was calculated using Johnson and Orme's (Johnson and Orme 2010) formula, which was specifically designed for choice-based conjoint analyses. The sample size was determined using the Johnson-Orme formula (Johnson and Orme 2010) for choice-based conjoint analyses. While the calculated minimum sample threshold was 42 participants, we selected a substantially larger sample size to enhance statistical power. Participants were recruited from Germany through the online panel Prolific between November 11-12, 2024. German language proficiency was established as an inclusion criterion since the survey was conducted in German. Furthermore, participants were required to be at least 18 years of age. The initial dataset comprised 323 participants. Following a thorough data cleaning procedure, which involved the exclusion of eight participants due to incomplete responses, the final sample comprised 315 participants (see Table 2 for detailed sociodemographic characteristics). Participants were remunerated for their participation through the Prolific platform.

4 Results

4.1 Overview of analysis approach

Based on the preregistered analysis plan, we employed a conventional conjoint approach. As this analysis did not reveal user preferences for specific attributes, we conducted additional, explorative cluster analyses in order to test for the presence of heterogeneous preferences within the sample population. This observation prompted the implementation of an explorative analysis with a segmentation approach to identify and examine potential heterogeneity in preferences. The analysis utilised a two-stage methodology: initially, a latent class analysis (LCA) employing a Gaussian mixture model with two components to identify distinct preference patterns, followed by logistic regression for each identified cluster. The analysis examined four key attributes: explainability, reliability, AI certification seal, and fairness. For each segment, the approach generated probability-based preference scores across attribute combinations, allowing the identification of segment-specific preferred combinations while considering the interaction effects between attributes.

Table 1. Logistic regression.

	Total sample. (n = 315)	Cluster 1. (n = 182)	Cluster 2. (n = 133)
Explainability	0.0042	0.1447***	−0.1899***
Reliability	0.0183	−0.0669	0.1371*
AI certification seal	0.0111	0.1926***	−0.2395***
Fairness	0.0155	−0.2692***	0.4090***

Results show regression coefficients (β) that indicate attribute preferences within each cluster based on logistic regression analysis. Positive coefficients indicate positive preferences for an attribute, while negative coefficients indicate negative preferences. Statistical significance determined by z-tests. P-values indicate significance levels: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

4.2 Hypothesis testing results

The analysis of the complete sample revealed no significant preferences across the four key attributes explainability, reliability, AI certification and fairness (see Table 1). For the analyses including the whole sample, all hypotheses need to be rejected.

4.3 Research question: most preferred attribute combinations

The Choice-Based Conjoint Analysis of the total sample revealed a consistent preference pattern, with the most preferred combination achieving a 50.8% selection probability. The optimal combination consisted of no explainability, medium reliability (65%), no AI certification seal, and high fairness. The observed similarity in the range of preference probabilities (50.3% to

Table 2. Sample size and description.

	Total sample. (n = 315)	Cluster 1. (n = 182)	Cluster 2. (n = 133)
Women	158 (50.16%)	86 (47.3%)	67 (50.4%)
Men	153 (48.57%)	92 (50.5%)	66 (49.6%)
Diverse	3 (0.95%)	3 (1.6%)	–
Not Specified	1 (0.32%)	1 (0.5%)	–
Age	18–92 (M = 32.6; SD = 10.7)	18–92 (M = 32.5; SD = 10.4)	18–92 (M = 32.9; SD = 10.1)
University Degree	183 (58.1%)	103 (56.6%)	80 (60.2%)
Highest School Degree	71 (22.5%)	44 (24.2%)	27 (20.3%)
University Entrance Qualification	61 (19.4%)	35 (19.2%)	17 (12.8%)
AI Acceptance	M = 7.27 (SD = 1.75)	M = 7.18 (SD = 1.74)	M = 7.40 (SD = 1.75)
AI Fear	M = 4.65 (SD = 1.83)	M = 4.79 (SD = 1.90)	M = 4.46 (SD = 1.69)

50.8%) across the top combinations suggests the presence of relatively uniform preferences within the overall sample.

4.4 Explorative analysis

4.4.1 Cluster analysis results

We employed Latent Class Analysis (LCA), a special type of finite mixture model for clustering with discrete variables, rather than k-means. LCA has been demonstrated to be advantageous for our Choice-Based Conjoint data, as it has been shown to directly model choice probabilities and to better handle categorical attributes. In the context of model selection, a comparative analysis was conducted between BIC and AIC values across a range of 1–5 cluster solutions. The 2-cluster solution was identified as optimal, exhibiting both significant improvement over a single cluster ($\chi^2 = 83.2$, $p < 0.001$) and high entropy (0.82), indicative of clear segment separation. The LCA identified two distinct participant clusters ($n_1 = 133$, $n_2 = 182$), each demonstrating significantly different preferences for AI system attributes. The analysis revealed notable differences in both preference patterns (see Table 1) and demographic characteristics between the identified clusters, providing more actionable insights than the initial overall analysis.

4.4.2 Hypothesis testing results for both clusters

Separate logistic regression analyses were conducted for each cluster to test our hypotheses, with choice as the dependent variable and attribute levels as the independent variables. Statistical significance was assessed using z-tests ($p < 0.05$). The model showed good fit with Pseudo R^2 values of 0.01027 (Cluster 1) and 0.01900 (Cluster 2), and highly significant likelihood ratio tests ($p < 0.001$).

In the context of the study, hypothesis H1 postulated that individuals would prioritise reliability over fairness and explainability. However, the analysis revealed that this hypothesis is not supported in both clusters. Cluster 1 exhibited no significant effect for reliability ($\beta = -0.067$, $z = -1.344$, $p = 0.179$) and a negative fairness effect ($\beta = -0.269$, $z = -5.409$, $p < 0.001$). Cluster 2 demonstrated significant effects for both reliability ($\beta = 0.137$, $z = 2.341$, $p = 0.019$) and fairness ($\beta = 0.409$, $z = 6.980$, $p < 0.001$), with fairness showing stronger influence.

The second hypothesis, which posited that trust seals would be more effective than explanations, received mixed support across clusters. Cluster 1 exhibited stronger preferences for AI certification ($\beta = 0.193$, $z = 6.310$, $p < 0.001$) compared to explainability ($\beta = 0.145$, $z = 4.742$, $p < 0.001$), thereby providing support for the hypothesis. Conversely, Cluster 2 exhibited adverse preferences for both attributes, with AI certification demonstrating a more pronounced negative effect ($\beta = -0.240$, $z = -6.660$, $p < 0.001$) compared to explainability ($\beta = -0.190$, $z = -5.285$, $p < 0.001$).

The third hypothesis postulated that individuals would demonstrate stronger preference for communicated trustworthiness (AI seal) over actual technical trustworthiness (reliability and fairness). This hypothesis received partial support. Cluster 0 exhibited stronger preferences for communicated trustworthiness through AI certification ($\beta = 0.193$, $p < 0.001$) compared to technical trustworthiness indicators. In contrast, Cluster 1 demonstrated stronger inclination toward technical trustworthiness indicators (fairness: $\beta = 0.409$, $p < 0.001$; reliability: $\beta = 0.137$, $p = 0.019$) over AI certification ($\beta = -0.240$, $p < 0.001$).

4.4.3 Research question: most preferred attribute combinations for both clusters

The results of the first cluster demonstrated a clear preference structure, with the most preferred combination achieving a 62.4% selection probability. This optimal combination comprised of no explainability, high reliability (99%), no AI certificate, and limited fairness. The results of the second cluster revealed different preferences, with the most preferred combination attaining a 66.9% selection probability. This optimal combination was characterised by its low explainability, medium reliability (65%), low AI certificate, and high fairness.

4.4.4 Demographic and attitudinal differences between clusters

Further statistical analysis revealed significant demographic differences between the clusters. Statistical analyses were performed using independent samples t-tests for continuous variables, while chi-square tests were employed for categorical variables. The t-tests revealed significant differences between the clusters in AI acceptance ($t = -6.453$, $p < .001$) and AI fear ($t = 9.701$, $p < .001$), as well as age ($t = -2.147$, $p = .032$). Cluster 1 reported higher AI acceptance and lower AI anxiety, whereas cluster 2 reported lower AI acceptance and higher AI anxiety. While age differences were small, Cluster 2 participants were slightly older than Cluster 1 participants. Detailed demographic characteristics for both clusters can be found in Table 2. Educational backgrounds differed significantly between clusters ($\chi^2 = 86.289$, $p < .001$). The distribution of educational qualifications varied notably, with Cluster 1 showing a higher proportion of university degrees, while Cluster

2 had a higher percentage of participants with highest school degrees and university entrance qualifications. Gender distribution also showed significant variation between clusters ($\chi^2 = 111.259$, $p < .001$). While Cluster 1 demonstrated an almost equal distribution between female and male participants, Cluster 2 showed a more diverse pattern, including small percentages of participants identifying as other gender or preferring not to specify. Comprehensive descriptive statistics for both clusters are presented in Table 2.

5 Discussion

This study examined user preferences regarding key attributes of AI systems, with a particular focus on the relative importance of explainability, reliability, AI certification, and fairness in users' decisions to engage with AI systems. The findings of the study revealed a more complex picture than initially hypothesised, with no attributes standing out when analysing the whole sample. Cluster analyses, however, revealed that there are distinct user groups with opposite preferences. These results offer significant implications for both the theoretical understanding of AI system development and the practical applications of these systems.

5.1 Heterogeneous user preferences and trust development

The lack of significant overall preferences across the entire sample is notable, as it suggests that the preferences of different user groups effectively balance each other out. However further analysis, employing cluster analysis, revealed two distinct clusters of users with contrasting preferences, challenging the assumption of uniform user priorities in AI system adoption. Cluster 1 demonstrated strong preferences for AI certification and explainability while expressing negative preferences for fairness. In contrast, Cluster 2 prioritised fairness and reliability while showing negative preference towards choosing explainability and AI certification. This group demonstrated higher AI acceptance and lower AI fear scores, indicating a more confident approach to AI interaction.

These findings contradict our first hypothesis that users would universally prefer reliability over fairness and explainability. Instead, we found that reliability's importance varies significantly between user groups, with Cluster 2 showing moderate preference for reliability while Cluster 1 showed no significant effect. This challenges previous findings by Kaplan et al. (2023) who identified reliability as the primary predictor of trust in AI systems. Our results suggest that the relationship between system reliability and user preference may be more nuanced and influenced by individual differences in AI attitudes.

5.1. Understanding versus trust: the role of explainability and certification

An analysis of the relationship between explainability and trust signals (H2) across the entire sample revealed no significant overall preferences, consistent with our findings regarding other attributes. However, distinct patterns emerged at the cluster level, indicating that while users lack universal preferences, individual groups demonstrate clear inclinations towards or against both mechanisms.

The findings of the present study showed that preferences for transparency (i.e. explainability) and trust signals (i.e. AI certification seals) are interconnected. These results contradict the hypothesis (H2) that users would prioritise one over the other. Cluster 1 exhibited a preference for both explainability and certification, indicating that these users seek diverse forms of external validation when evaluating AI systems. This finding is consistent with theoretical frameworks suggesting that users may employ multiple trust-building mechanisms when dealing with complex systems (Paaß and Hecker 2020).

In contrast, Cluster 2 demonstrated negative preferences for both explainability and certification while favouring fairness and reliability. This pattern challenges previous assumptions about users universally needing system understanding or validation. Whilst an initial interpretation indicated that users might establish trust through alternative means, such as direct system performance observation, a more nuanced interpretation is required. It is crucial to acknowledge that the present study measured system choice as opposed to trust directly, as these are related yet distinct concepts. This perspective aligns with the findings reported by (Kieslich, Keller, and Starke 2022) and (Bao et al. 2022), which suggest that some users may exhibit indifference towards specific ethical considerations of AI systems, particularly transparency mechanisms. In contrast to actively seeking diverse trust pathways, these users may prioritise performance-oriented attributes with immediate practical impact (e.g. fairness and reliability), while demonstrating comparatively less concern for the justification of system decisions or for external validation. This finding extends the observation made by Ferrario and Loi (Ferrario and Loi 2022) regarding the potential of increased system familiarity to reduce reliance on conventional trust mechanisms. The results suggest that user preferences for AI system attributes are more intricate and diverse than previously understood.

The parallel preferences for or against both transparency and trust mechanisms across clusters suggest that these attributes may be more interconnected than previously theorised, potentially representing different aspects of a single underlying approach to AI system evaluation, potentially relying on external communication about the system.

5.2. Technical versus communicated trustworthiness

Analysis of preferences between technical and communicated trustworthiness (H3) revealed no significant trends across the complete sample. However, distinct patterns emerged at the cluster level, demonstrating how different user groups approach trust development in contrasting ways. Our third hypothesis regarding the preference for communicated trustworthiness over technical trustworthiness revealed a more complex dynamic than anticipated. The analysis showed a clear division between clusters in how they evaluate and prioritise different forms of trustworthiness.

Cluster 1's stronger preference for AI certification aligns with Knowles and Richards' (Knowles and Richards 2021) argument about the importance of regulatory frameworks in building public trust. The fact that this group showed no significant preference for reliability, traditionally considered a cornerstone of technical trustworthiness, further supports the interpretation that they rely more heavily on external validation than direct performance assessment.

In contrast, Cluster 2's clear preference for technical trustworthiness indicators (fairness and reliability) over certification presents an unexpected pattern that challenges conventional wisdom about trust development. This group appeared to reject communicated trustworthiness in favour of direct performance indicators, suggesting that some users might be more critical of external validation mechanisms, preferring instead to evaluate systems based on their actual performance characteristics.

These contrasting preferences raise important questions about the conventional approach to AI system deployment, which often emphasises standardised certification and documentation practices. Our results suggest that different user groups might require fundamentally different approaches to establishing trustworthiness. This has significant implications for our research question about attribute combinations, as it indicates that the effectiveness of various trust-building features might depend heavily on users' underlying approach to technology evaluation. The finding challenges the assumption that adding more trust signals universally enhances system trustworthiness, suggesting instead that targeted combinations might be more effective for specific user groups.

5.3. Preferred attribute combinations and their implications

The Choice-Based Conjoint Analysis of the total sample revealed a consistent preference pattern, with similar selection probabilities (50.3% to 50.8%) across top combinations. The combination that was selected most frequently, achieving a 50.8% selection probability, featured no explainability, medium reliability (65%), no AI certification seal, and high fairness. This relative uniformity in preference probabilities suggests balanced preferences within the overall sample. However, cluster analysis revealed more nuanced patterns.

Cluster 1's optimal combination revealed an unexpected preference pattern: they favoured high reliability without explainability or fairness features. This finding suggests a minimalist approach that focuses solely on technical performance. This pattern might indicate that for some users, system complexity itself could be a source of concern, leading them to prefer simpler, more straightforward implementations when given direct choice scenarios. This interpretation aligns with Springer and Whittaker's (Springer and Whittaker 2020) observation that some users actively avoid detailed system information, fearing that increased knowledge about data processing might discourage future use.

Cluster 2's preferred combination presents an equally unexpected but different pattern: they opted for systems with low explainability, moderate reliability (65%), low AI certification, but high fairness. This preference pattern is particularly significant as it challenges multiple assumptions about AI system design. Their strong emphasis on fairness while accepting lower levels of other attributes supports Sullivan et al.'s (2022) findings about the growing importance of ethical considerations in user trust development. The willingness to accept moderate reliability when paired with high fairness suggests that users might be making sophisticated trade-offs between technical performance and ethical considerations.

This finding extends beyond simple feature preferences to suggest deeper differences in how users conceptualise and evaluate AI trustworthiness. While system designers often strive to maximise all positive attributes, our results indicate that different user groups might have fundamentally different visions of what constitutes an ideal AI system. This has significant implications for AI system development, suggesting that optimising for all features simultaneously might actually make systems less appealing to certain user groups who prefer more focused or streamlined implementations. These patterns also suggest an important refinement to theories about trust development in AI systems. Rather than treating trust as a cumulative product of positive features, our findings indicate that users might employ different strategic approaches to system evaluation, with some preferring focused, performance-oriented systems while others seek more balanced implementations that prioritise ethical considerations over maximum technical performance.

5.4. Limitations and future research

Despite the fact that the methodological approach of presenting isolated attribute combinations may appear to be abstracted from real-world AI interactions, this design choice offers crucial advantages. As Orme (Orme 2010) observes, conjoint analysis allows for the precise measurement of relative preferences for specific AI system attributes by deliberately reducing complexity and controlling for confounding variables. This approach enables the observation of realistic decision-making processes through forced trade-offs. Whilst the present controlled experimental setting may appear to be detached from reality, it offers unique insights that would be difficult to obtain from studying complete AI systems where multiple factors interact simultaneously.

Nevertheless, several limitations should be considered when interpreting these results. First, our study focused on a specific healthcare decision-making scenario, and findings may not generalise to other AI applications. Second, while our sample was diverse in terms of age and education, it was limited to German participants and may not be fully representative of all potential AI system users. A final limitation of the present study concerns the abstract representation of system attributes. While these standardised presentations are necessary for the isolation of attribute effects, they may not fully capture real-world user experiences with AI systems, potentially resulting in a limitation of ecological validity. It is recommended that future studies build upon the present approach by incorporating research that employs more concrete, interactive representations of explainability features through functional prototypes or simulations that better approximate real-world AI interactions.

A significant consideration for future research is the role of AI literacy in shaping user preferences and trust development, building on recent work on digital literacy and AI adoption. The stark differences between our identified clusters align with previous findings (Cox 2024) suggesting that varying levels of AI literacy significantly influence how users approach and evaluate AI systems. Future studies should explicitly measure AI literacy to examine whether it mediates the relationship between system attributes and user trust. This could help explain why some users prioritise technical performance while others rely more heavily on external validation. Future research could also explore how these preference patterns manifest in different AI application contexts, and how they might be influenced by cultural and organisational factors. The development of adaptive trust-building mechanisms that can accommodate different user groups remains an important avenue for investigation. Additionally, longitudinal studies examining how trust development patterns evolve with increased AI exposure and literacy would provide valuable insights for system design and implementation.

6 Conclusion

The present study examined the development of user trust in AI systems, revealing four crucial insights. Firstly, the findings contradict the prevalent assumption that a uniform pattern exists; instead, we observed contrasting preferences among user groups that cancel each other out. Secondly, the present study found that user preferences regarding understanding (explainability) and trust signals (AI certification seal) are coupled. This finding challenges the initial research question about whether users prefer one approach over the other. Instead of a clear preference for one over the other, a distinct group of users exhibited a desire for both mechanisms, suggesting a balanced approach to trust and transparency. Thirdly, distinct user clusters with contrasting trust-building strategies were identified: one group relies heavily on both external validation and transparency, exhibiting higher AI anxiety; in contrast, another prioritises externally communicated reliability aspects and fairness metrics and shows greater AI acceptance. These results raise important questions about the effectiveness of current transparency approaches in reaching all user populations, and the potential need for alternative strategies to engage users who demonstrate less interest in critically examining AI systems beyond their direct performance outcomes. The methodological approach employed in this study has exposed sophisticated tradeoffs in user preferences, particularly regarding the prioritisation of technical performance over ethical considerations. Users comfortable with AI technology often accept moderate reliability when paired with high fairness, while AI-anxious users place greater emphasis on transparency measures and certification seals. The findings demonstrate that effective AI integration requires a shift from universal design principles to adaptive approaches that acknowledge combined preferences and adapt to diverse user groups.

Acknowledgements

This work has been partly supported by the Research Center Trustworthy Data Science and Security (https://rc-trust.ai/), one of the Research Alliance centers within the UA Ruhr (https://uaruhr.de). We sincerely appreciate their support. Additionally, generative AI tools, specifically Claude 3.7 Sonnet, were used during the writing process to support the linguistic refinement and structural revision of texts originally authored by the researcher, including text formulation, proofreading, and debugging of code segments

Author contributions

CRediT: **Nur Efsan Cetinkaya:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft; **Nicole Krämer:** Conceptualization, Funding acquisition, Methodology, Resources, Supervision, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by Research Center Trustworthy Data Science and Security.

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